

MSc Semester Project 1

Field	Description
Project focus	AI-Driven Spam Detection and Scam Engagement
Document purpose	Assignment brief ready for direct distribution to students
Structure	Background, learning goals, tasks for students, and expected deliverables

1. Background

Email spam, particularly advance-fee fraud such as the well-known "Nigerian Prince" scam, remains one of the most persistent threats in online communication. These messages are designed to mislead recipients into sending money, sharing personal information, or continuing fraudulent conversations under false pretences.

This assignment explores how artificial intelligence and machine learning can be used to detect such malicious emails automatically. In addition to classification, the assignment introduces a generative AI component that can produce realistic replies to scam messages. The purpose of this second component is not to assist scammers, but to keep them occupied with fabricated conversations so that their time and effort are diverted away from real victims.

The first stage of the work focuses on building or integrating a model capable of classifying incoming email text as legitimate or spam, with special emphasis on scam-style messages. Once a message is flagged as suspicious, a generative AI tool can be used to draft context-aware responses and maintain multi-turn interaction if further scam emails are received.

Through this assignment, participants will gain practical exposure to text classification, feature extraction, prompt design, conversational AI, and the ethical boundaries of automated engagement systems. The assignment combines defensive filtering with proactive disruption, creating a useful case study at the intersection of cybersecurity and AI.

2. Learning Goals

- Reduce the likelihood that users will engage with fraudulent scam emails.
- Use AI-generated responses to occupy scammers and limit their ability to target real victims.
- Give students practical experience with machine learning classification and controlled generative AI workflows.

3. Tasks for Students

1. Create or collect a simulated dataset of scam emails, with emphasis on "Nigerian Prince"-style fraud scenarios. The project should operate only on email body text and should avoid using real personal data.

2. Develop a preprocessing pipeline that parses incoming email text and extracts useful information for classification, such as keywords, sentence patterns, named entities, or message structure.
3. Review related work on spam detection, phishing classification, and AI-assisted email filtering in order to justify the chosen method.
4. Design, train, or integrate an AI/ML model that classifies emails as legitimate or spam. Explain the selected approach, feature choices, and evaluation method.
5. Leverage a generative AI model, such as ChatGPT, Claude, or another comparable system, to produce automatic replies to simulated scam emails.
6. Demonstrate multi-turn behavior by showing how the system can continue responding when additional scam messages are received.
7. Discuss safety, legal, and ethical issues associated with automated engagement, including the risks of escalation, data handling, and unintended misuse.
8. Prepare a short report summarising the dataset, methodology, system design, results, limitations, and suggested future improvements.

4. Expected Deliverables

- A concise written report in academic style.
- Evidence of implementation, experimentation, or policy design, depending on the selected project.
- A short discussion of limitations, risks, and possible future improvements.
- References to any tools, datasets, standards, or sources used during the work.